

Business Analytics

① Business Analytics: Involves the iterative and methodical approaches in inspecting or investigating business data, by applying statistical models and quantitative methods to ~~solve~~ help businesses cut cost and solve problems.

② Business Intelligence: Is actionable insights ^{from raw data} processed with tools and tech, that tells us where the business, how its performing, and which logical steps ~~that~~ should take that are tactical and strategic in nature.

③ BIG DATA: Large and complex information or datasets that are commonly characterised by, their variety, velocity, and high volume. They typically ~~may~~ require specialized technology to process.

④ DIAGNOSTIC ANALYTICS: The process analysing data to determine trends, root causes, and any correlation or events.

⑤ Prescriptive Analytics: An advanced form of analysis that recommends task or actions that need to taken to achieve a result in a given situation or scenario.

⑥ DATA ANALYTICS: A universal discipline of analysing ~~information~~ ~~to make raw data~~, to make conclusions the information. It often includes inspecting, cleaning, transforming and modeling data.

⑦ DATA DRIVEN DECISION MAKING: The routine way of basing strategic business decisions on data analysis rather than observation or intuition.

⑧ DESCRIPTIVE ANALYTICS: This is the interpretation of historical data to analyse trends and patterns as well as the current state of a business.

⑨ Predictive Analytics: The use of statistical methods and historical data to forecast future events

⑩ DATA LITERACY; The ability to read, understand and communicate with data. The ~~data~~ display of competency in understanding information from data analytics

Real world examples

- Business intelligence - A retail store using dashboards to monitor sales
- Big Data: Netflix processing millions of user interactions
- Diagnostic Analytics: Running ~~root~~ cause analysis on Takalot's sudden conversion drop
- Prescriptive Analytics: Vermeer logistics using algorithmic data to determine which are the efficient routes for their trucks to use
- Data driven decision making: A KFC marketing manager doubling on a campaign after data confirms sales uplift without increasing the cost.
- Descriptive data: ^{HK} HOD analysing an annual report to determine which department experienced the highest number of resignations
- Predictive Analytics: KPMG/PWC using past customer spending data and credit scores to see which customer has the highest probability on defaulting on loan repayments
- Data literacy: ~~Any~~ Non-technical project manager at J&T fashion learns to interpret sales data and successfully communicates with logistics.

STRUCTURED DATA: This information that is predefined and organised into tables with row and columns. This helps both humans and machines. An example would be an e-commerce platform maintains a product database where every item is assigned attributes like price, SKU, stock count.

DATA WAREHOUSE: It is a centralized repository that aggregates data from different operational systems like CRMs or ERP into a single storage unit that is optimised for querying and analytics. A large retail chain like Mr. Price consolidates from various in-store point of sale systems or online.

DATA VISUALISATION: The practice of transforming complex datasets into graphic visuals such as charts, graphs, maps, interactive diagrams, to convert data into understandable insights. Tools that achieve this are Power BI, Tableau, Google Looker.

Metric: Is quantifiable measurement, defined by a consistent formula used to track performance of a specific business activity such as sales or staff performance. An insurance company might track monthly recurring revenue or sales department monthly sales performance.

measure

~~Benchmark~~: An ~~industry~~ standard that ~~organisation~~ ^{measures} ~~itself~~ ^{aga} ~~products~~, services or operations against

BENCHMARK: An industry standard that an organisation measures itself against; comparing its procedures, products or services to identify possible area of improvement over time. H&M might measure the speed of order fulfillment and development products that help in increasing it.

Unstructured Data: This is the opposite of structured data; it lacks predefined model and is unorganised such email, videos, social media, or audio files. A digital security firm uses machine learning to process unstructured data by scanning ^{the} ~~test~~ content of emails to identify and filter phishing.

DATA MINING: ~~A statistical~~ A process of ~~search~~ using statistical algorithms and computational tools to extract hidden patterns, anomalies or correlations in large and complex data. ~~A data in~~ A bank might use data mining to analyze credit card transaction to detect fraudulent activity.

DASHBOARD: A visual interface that displays multiple consolidated ~~data~~ ~~multiple~~ data points, metrics, KPI's into a single view, to provide a snapshot of a status report or performance. A marketing manager uses dashboards to track campaign performance.

KEY PERFORMANCE INDICATORS (KPIs):

A specific, prioritized metric that is tied to a strategic business goal or objective. This serves as a key marker that measures progress toward the goal. For a business like Superbalist a business would measure "Cart abandonment Rate". This is tied to how effectively the company converts

SEGMENTATION: An analytical method of dividing broad datasets into smaller unique subset of data base on specific criteria like demographics, behaviour, or psychographics. Cotton on would group buyers/customers into "frequent high spenders" vs occasional shoppers, this can be used to target each group with segment specific loyalty rewards.

CUSTOMER & SALE KPIs

① **CONVERSION RATE:** A percentage of users that perform a desired action like clicking on an ad vs the number of people who were exposed to the same ad creative. E.g. Rep send 10000 mailers to customer/subscribers but only 250 click on it.

② **CUSTOMER LIFETIME VALUE:** This is a total estimated profit that a company expects from a particular client over the entire relationship with the client/business. ~~Levante~~ A subscription based software might calculate that a customer with a monthly gross contribution of \$16000 and a 2.5% churn rate yields

a lifetime value of \$1 600 000. This informs the business on how much they are willing to spend acquisition.

CUSTOMER RETENTION RATE: A metric that measures the percentage of customers that a business keeps over a specific period calculated like this

$$CRR = \frac{\text{No. of customers} - \text{new acquisitions}}{\text{Count at the start}}$$

$$CRR = \frac{100 - 10}{100} =$$

$$CRR = 90\%$$

AVERAGE ORDER VALUE: A snapshot that shows you the customer average spend

A snapshot of customer spending habits that tracks the average customer spend at each time a customer places an order. An e-commerce store would increase AOV by 10%-15% by upselling at check-out

CUSTOMER ACQUISITION COST: The total cost including sales outreach, marketing, advertising which is required to acquire a new customer. If marketing spends R1000 to acquire 100 new paying customer, then they spent R100/per new paying customer.

CHURN RATE : The number of paying / subscribed customers who discontinued their relationship with the company over a specific period. A telecoms company might find a 3% churn rate and need to investigate the quality of service delivery, pricing or value of product offering.

NET PROMOTER SCORE (NPS) : A statistical method measuring customer satisfaction and loyalty, by asking how likely they are to rate recommend the business between 1 - 10.

SALES FUNNEL : A visual representation of a customer journey tracking the initial awareness through to the final stage of purchase. A marketing department might analyze that while 40% of visitors view products only 2.5% purchase to pin point friction points and resolves.

MARKETING & CAMPAIGN ANALYTICS

MARKETING CAMPAIGN : A coordinated series of activities designed to achieve specific business goals within a specified period of time. An example could new product launches by Apple, or even tech and software updates Apple did the "Shot on iPhone" which aimed to showcase Apple's camera quality.

Incremental Revenue: The ~~additional~~ additional revenue a business generates over the period of a campaign, calculated by sales lift during the campaign against a baseline. A business might use predictive analytics in capturing new business instead of allocating budget to the existing consumer base.

RETURN ON AD SPEND: A metric used to measure the gross revenue of and spent on advertising. This is calculated as $\frac{\text{total campaign revenue}}{\text{total ad spend}}$.

BOUNCE RATE: The percentage of visitors who leave the website after visiting one page. This often is a signal of poor technical performance or lack of relevance. A marketer might see a big drop off from landing page and deploy A/B testing of different headlines and/or loading speeds.

LEAD GENERATION: The process of capturing ~~customer~~ contact information in exchange for value like a masterclass, discount or promo code. A B2B company might run a webinar where users sign-up with their emails in exchange for industry insight.

POST CAMPAIGN ANALYSIS: A systemic analysis of a campaign measured against predefined business goals. Real world application involves forecasting and budget allocation

RETURN ON INVESTMENT: This is a broad financial metric used to assess profitability, by measuring net profit vs cost incurred. ROAS focuses on Ad efficiency and ROI on overall business health

CLICK THROUGH RATE: ~~The number~~ The ratio or percentage of users who click through on an ad or link vs the amount exposed to it. A marketer might find that a testimonial video performs better than a static image, This guides all future content planning

A/B TESTING: A comparison of two marketing assets such as an email headline or website button to see which one statistically performs better. An e-commerce site might test out two check out pages and measure which one performs better

MARKETING ATTRIBUTION: An analytical science that measures which touch point such as search ad, social media clicks or email newsletters deserves credit for converting.

References

1. Sales force
2. Rocketagent
3. Improvado
4. Mercury
5. Upgrowth